

Near-Unity Singlet Fission on a Quantum Dot Initiated by Resonant Energy Transfer

Jie Zhang, Hayato Sakai,* Katsuaki Suzuki, Taku Hasobe, Nikolai V. Tkachenko, I-Ya Chang, Kim Hyeon-Deuk, Hironori Kaji, Toshiharu Teranishi,* and Masanori Sakamoto*

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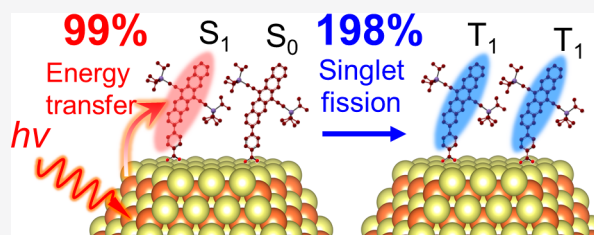


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Supporting Information

ABSTRACT: The conversion of a high-energy photon into two excitons using singlet fission (SF) has stimulated a variety of studies in fields from fundamental physics to device applications. However, efficient SF has only been achieved in limited systems, such as solid crystals and covalent dimers. Here, we established a novel system by assembling 4-(6,13-bis(2-(triisopropylsilyl)ethynyl)pentacen-2-yl)-benzoic acid (Pc) chromophores on nanosized CdTe quantum dots (QDs). A near-unity SF ($198 \pm 5.7\%$) initiated by interfacial resonant energy transfer from CdTe to surface Pc was obtained. The unique arrangement of Pc determined by the surface atomic configuration of QDs is the key factor realizing unity SF. The triplet–triplet annihilation was remarkably suppressed due to the rapid dissociation of triplet pairs, leading to long-lived free triplets. In addition, the low light-harvesting ability of Pc in the visible region was promoted by the efficient energy transfer ($99 \pm 5.8\%$) from the QDs to Pc. The synergistically enhanced light-harvesting ability, high triplet yield, and long-lived triplet lifetime of the SF system on nanointerfaces could pave the way for an unmatched advantage of SF.



INTRODUCTION

Singlet fission (SF) in organic chromophores is an exciton multiplication process in which a spin-singlet excited state splits into a pair of spin-triplet excited states.^{1–6} The conversion of a high-energy photon into two photons/excitons by SF has stimulated a variety of studies in fields from fundamental physics to solar energy conversion.^{7–11} Because SF is an intra- and intermolecular phenomenon that requires suitable intermolecular electronic coupling, investigations of SF have focused on organic materials with highly dense packing, such as crystalline solids, aggregates, and extremely concentrated solutions.^{12–16} In addition, covalently tethered dimers have been developed to optimize chromophore configurations for SF.^{17–22} Earlier reports of SF suggested that controlling for intermolecular orientation and/or distance holds the key to achieving efficient SF.^{22–27} Exploring an ideal system exhibiting a favorable molecular arrangement for SF is still challenging and strongly demanded.

An inorganic quantum dot (QD) surface is a promising platform for concentrating functional molecules in a limited region.^{28,29} QDs present an excellent light-harvesting property that can be systematically controlled by the size-dependent quantum confinement effect, providing intense absorptions and emissions ranging from the ultraviolet (UV) to the infrared (IR) region.^{30–32} The versatile charge/energy transfer between QDs and surface-attached molecules suggests that QDs promote the light-harvesting ability of a whole system.³³ Therefore, composites of QDs and organic molecules can be

an ideal system for establishing efficient SF. However, a favorable sensitization system (Förster resonance energy transfer (FRET) or Dexter-like singlet energy transfer) for a QD–molecule composite is still under debate.^{34,35} Furthermore, the triplet yields by SF on previously reported QD–molecule systems have been far from the ideal value.^{34,35} Even for the QD–molecular dimer systems employing intra-molecular SF, the triplet yield was 160%.³⁴ The triplet yield of the QD–molecular monomer systems (intermolecular SF system) (113%) initiated by a Dexter-like energy transfer is well below the ideal value.³⁵ The construction of ideal systems based on an understanding of the underlying mechanism and key factors has been strongly required.

Here, we successfully demonstrate an ideal composite for SF by assembling 4-(6,13-bis(2-(triisopropylsilyl)ethynyl)pentacen-2-yl)benzoic acid (Pc) molecules on the surface of CdTe QDs. From the results of femtosecond transient absorption (fsTA) measurements, the triplet yield of the intermolecular SF system was determined to reach $199 \pm 7.6\%$, which is the highest yield among the QD–molecule systems. Furthermore, the total triplet yield of SF via FRET was $198 \pm$

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5.7% due to unity energy transfer between Pc and the QDs ($99 \pm 5.8\%$). Furthermore, the recombination of triplet pairs was remarkably suppressed in the current system. These results suggested that the excitons generated in CdTe can be transferred to Pc and completely separated into two triplet excitons. A double-quantum (DQ) experiment using dynamic nuclear polarization enhanced nuclear magnetic resonance (DNP-NMR) and a computational simulation unveiled the unique molecular arrangement of Pc dimer on CdTe QDs, reflecting that the surface atomic configuration realized the efficient SF.

RESULTS

Material Design. A schematic of the QD–molecule composite and all the relevant photophysical processes is displayed in Figure 1a. CdTe QDs were selected as a scaffold

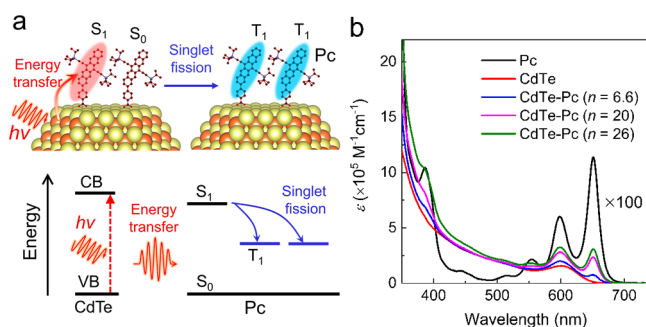


Figure 1. Schematic illustration and absorption spectra of the QD–molecule composite for SF. (a) Schematic illustration of SF at the CdTe–Pc heterointerface. An exciton generated in a CdTe QD transfers to Pc, forming a molecular singlet exciton. Then, the singlet excited Pc molecule undergoes SF with a neighboring molecule to produce two triplet excitons. (b) UV–vis–near-IR absorption spectra of CdTe QDs, CdTe–Pc, and Pc in toluene (n is the average number of Pc molecules on a single QD).

for the QD–molecule system. Pc derivatives have been extensively investigated as SF chromophores because they cause rapid and efficient SF due to well-matched energy levels, $2E(T_1) < E(S_1)$, where $E(T_1)$ and $E(S_1)$ are the energy levels of the lowest triplet and lowest singlet excited states, respectively.^{1,36} Near-unity triplet yields have been achieved in solids and solutions of Pc derivatives.^{13,37} However, the light-harvesting ability of Pc derivatives is relatively low because of their extremely low absorption coefficients (ϵ) ($[1\text{--}6] \times 10^3 \text{ M}^{-1} \text{ cm}^{-1}$ in the 400–500 nm range).³⁵ In contrast, the ϵ value of CdTe QDs is $\sim 2 \times 10^5 \text{ M}^{-1} \text{ cm}^{-1}$ in the same range,³¹ approximately 2 orders of magnitude higher than that of Pc. To achieve an efficient FRET from QDs to Pc, the emission of QDs was controlled to overlap with the absorption of Pc at approximately 600 nm. The low absorption of Pc is covered by the absorption of CdTe, which promotes the efficiency of exciton generation in Pc against incident photons. The excitons generated in the CdTe QDs can transfer to Pc through FRET. SF from the excited singlet state of Pc produces a maximum of two triplet states.

Synthesis and Characterization of Samples. CdTe QDs were synthesized by a hot injection method and dissolved in toluene solution (see Section 1 in the Supporting Information). Transmission electron microscope (TEM) images indicate that the size of the CdTe QDs is $3.5 \pm 0.3 \text{ nm}$ (Figure S3). X-ray diffraction (XRD) patterns indicate that

the crystal structure of the CdTe QDs is zinc blende (JCPDS 15-0770) (Figure S4). Pc molecules were assembled on QDs by a ligand exchange reaction. The carboxylic acid group attaches Pc to the CdTe surface. The average number of Pc molecules on a single QD (n) is quantitatively controlled by changing the concentration of Pc in the ligand exchange reaction. Three samples with different n values of 6.6, 20, and 26 were prepared. UV–vis–near-IR absorption spectra of CdTe–Pc show the characteristic absorption features of Pc (555, 600, and 650 nm) and CdTe (600 nm) (Figure 1b). The value of n was calculated spectroscopically by comparing the concentrations of Pc and CdTe in the CdTe–Pc toluene solutions (see Section 4 in the Supporting Information).

Femtosecond Transient Absorption Measurements.

fsTA measurements were conducted to observe the excited-state dynamics of CdTe QDs and CdTe–Pc. A two-dimensional (2D) fsTA map of the pristine CdTe QDs is displayed in Figure 2a. Upon excitation of the QDs by a 470 nm laser pulse, the 2D fsTA map is dominated by a strong negative signal at approximately 600 nm, which is attributed to the bleaching caused by the state filling of QDs induced by the transition of electrons from the valence to the conduction band.³⁸ The photoinduced absorption at approximately 540 nm is derived from a transition from an excited state to a higher excited state or a biexcitonic state.³⁹ The global fit (Figure 2b; see also Section 10 in the Supporting Information) gives the so-called Poisson component as dominating component and suggests that the QD excited state lifetime is 11 ns. The recovery of the ground-state bleaching is not complete within the time window of the fsTA system (5300 ps).

The SF in CdTe–Pc samples was first studied by selective excitation of Pc at 660 nm (Figures S21 and S22). Then, the excitation was switched to 470 nm, which predominantly excited the QD (>97%). Among the three CdTe–Pc samples, SF occurs most efficiently for the sample with $n = 26$. As shown in the fsTA spectrum of CdTe–Pc (Figure 2d), the recovery of the ground-state bleaching of QDs is $8.9 \times 10^9 \text{ s}^{-1}$, which is much faster than that of pristine QDs, suggesting that the excitons in the CdTe QDs are quenched by FRET to the surface Pc. The recovery of the bleaching is accompanied by two positive signals of excited-state absorption, which is characteristic of Pc in the singlet excited state (Pc (S_1)) (light green line region in Figure 2d). Since the rate of increase in Pc(S_1) matches the recovery rate of the ground-state bleaching of the CdTe QDs, excitons generated in the CdTe QDs are transferred to Pc(S_1) via FRET. The possibility of carrier transfer was eliminated by the absence of a TA signal of Pc radical cations or anions. Finally, the TA signal assigned to the triplet state (Pc(T_1)) was observed (yellow line region in Figure 2d). Since the decay rate of Pc(S_1) and growth rate of Pc(T_1) were similar to each other ($1.2 \times 10^9 \text{ s}^{-1}$), we believe that SF generates Pc(T_1). Although it is plausible that the direct triplet energy transfer at the CdTe–Pc interface can also form Pc(T_1), this process can be eliminated by the formation of Pc(S_1) before the formation of Pc(T_1). The time-resolved shift of TA confirmed that sequential FRET and SF occurred on CdTe–Pc.

To obtain more insight into the relaxation mechanism, time profiles at all wavelengths were generated by a global fitting analysis using a multiexponential function and the function based on Poisson distribution. Through the global analysis shown in Figure 2c, it is deduced that there are three

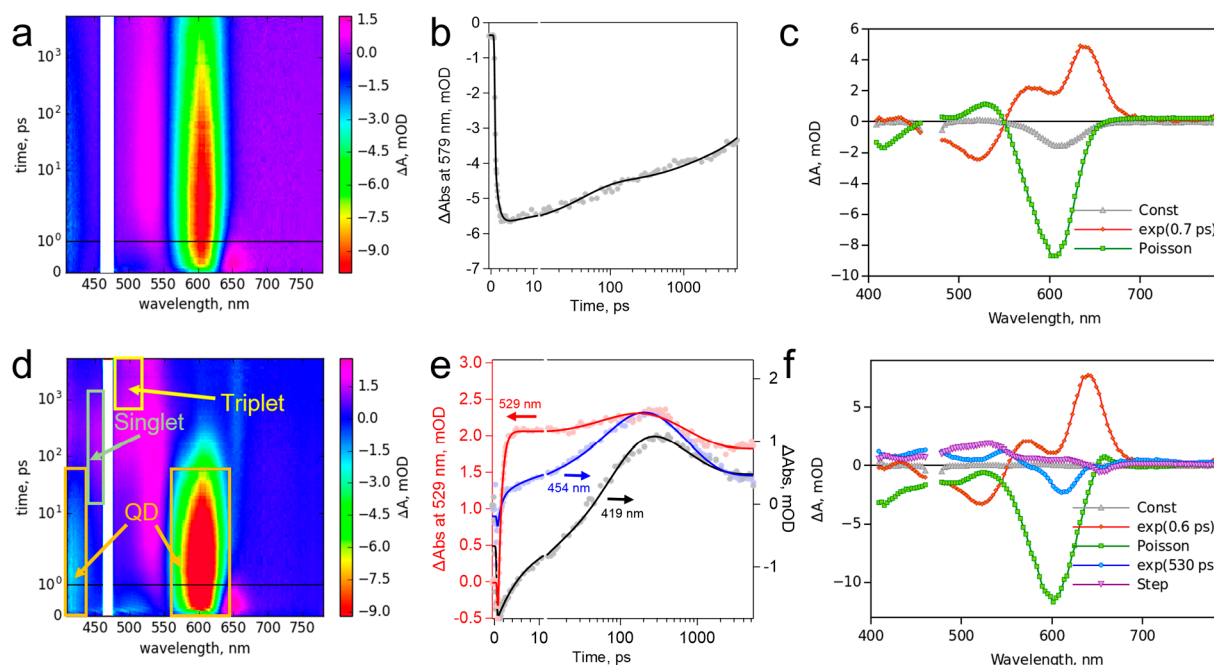


Figure 2. fsTA spectra of CdTe QDs and CdTe-Pc pumped at 470 nm. (a) 2D fsTA maps of CdTe QDs. (b) Kinetic profiles of CdTe QDs. The profiles are fitted by multiexponential growth functions and the function is described by a kinetic model assuming a Poisson distribution of adsorbates on quantum dots. The first component was extracted by exponential fitting, and the second component was extracted by the decay model utilizing the Poisson distribution. The solid line is the best fit. (d) 2D fsTA maps of CdTe-Pc ($n = 26$). Bleaching of QDs is quenched by energy transfer from CdTe to Pc (orange line region near 600 nm), accompanied by components of singlet Pc (light green line region near 454 nm) and triplet Pc (yellow line region near 529 nm). (e) Kinetic profiles of CdTe-Pc ($n = 26$). The profiles are fitted by multiexponential growth functions and the function is described by a kinetic model assuming a Poisson distribution of adsorbates on quantum dots. The first, third, and final components were extracted by exponential fitting, and the second component was extracted by the decay model utilizing the Poisson distribution. The solid lines are the best fits. (c) Global fitting analysis for the TA spectra in (a). Const. denotes the constant accounting for the sample emission, $\exp(0.7 \text{ ps})$. The component of the time constant of 0.7 ps is assigned to CdTe QDs by an exponential function. Poisson denotes the component of Poisson distribution. (f) Global fitting analysis for the TA spectra in (d). Const denotes the constant, $\exp(0.6 \text{ ps})$. The component of the time constant of 0.6 ps is assigned to CdTe QDs by an exponential function. Poisson denotes the component of Poisson distribution. The component of the time constant of 530 ps assigned to be $\text{Pc}(S_1)$ by an exponential function. Step denotes the component of the time constant of the long-lived lifetime assigned to be $\text{Pc}(T_1)$. Values in the inset are time constants.

evolution-associated spectra (EAS) with different lifetimes for the QDs. The component with a very short lifetime of 0.7 ps (red) is thermal relaxation of QDs to the lowest excited state. The EAS with a lifetime component of 90 ps (green), determined by the function based on a Poisson distribution,^{40–42} exhibiting the multistep recovery of the bleaching of the CdTe QDs correspond to the high-energy exciton state, exciton state, and defect trapping state, respectively.^{38,39} In the global fitting analysis shown in Figure 2f, the five EAS suggest the process of SF via FRET on CdTe-Pc. The EAS of CdTe-Pc with a lifetime component of 0.6 ps (red) is thermal relaxation of QDs, and the lifetime component of the decay model utilizing the Poisson distribution (green) is 90 ps. Although the quenching time constant is rather similar to that observed in CdTe QDs, this value is a different process because the concentration of quenchers in CdTe-Pc is 10 times greater than that of pristine QDs, which shows the energy transfer from QDs to Pc. Then, the EAS with a time constant of 530 ps (blue) is assigned to mainly the component of $\text{Pc}(S_1)$ with unique peaks at 454 and 540 nm. The next EAS with a time constant of $>5300 \text{ ps}$ (purple) corresponds to $\text{Pc}(T_1)$. This time-resolved shift of TA signals further identifies the occurrence of FRET and SF between Pc molecules on the surface of QDs. The quantum yield of FRET and the triplet yield of SF via FRET were calculated to be $99 \pm 5.8\%$ and $198 \pm 5.7\%$, respectively (see the Supporting Information for

details). As the triplet yield was the same as the value ($199 \pm 7.6\%$) of SF that occurred by selectively exciting only Pc with a 660 nm laser, the observed triplet states are generated by only SF. Therefore, the kinetic constant of SF was determined to be $1.2 \times 10^9 \text{ s}^{-1}$ from the TA spectra. The near-unity quantum yield of FRET could be ascribed to the excellent overlap between the emission of CdTe and the absorption of Pc. The calculated triplet yield of $198 \pm 5.7\%$ (SF efficiency: $99 \pm 5.8\%$) indicates that the CdTe-Pc composite could achieve SF with near-unity efficiency. The evaluation of the yields and the dynamics of other QD systems also used the same method. The near-unity SF efficiency of the current CdTe-Pc system was also supported by time-resolved fluorescence measurements (see Section 9 in the Supporting Information).⁴³

DISCUSSION

The importance of the intermolecular distance and morphology of SF chromophores for SF has been extensively demonstrated.^{23–25,44} The occurrence of SF requires a short intermolecular distance of Pc ($<1 \text{ nm}$) for intermolecular electronic coupling.^{36,37} To investigate the effect of the average intermolecular distance on the SF rate and triplet yield, we changed the intermolecular distance (i.e., surface Pc density) of the surface-attached Pc by changing n . The parameters expressing the average distance of Pc on CdTe and the

Table 1. Parameters of the QD-Pc Composites and Properties of SF

sample	<i>n</i>	density (nm ⁻²)	distance (nm)	FRET yield (%)	SF rate (ns ⁻¹)	triplet yield (%)
CdTe-Pc	6.6	0.31	1.8	87 ± 5.9	0	0
	20	0.95	1.0	95 ± 5.1	0.71	159 ± 6.8
	26	1.2	0.90	99 ± 5.8	1.2	198 ± 5.7
CdTe@CdS-Pc	30	0.65	1.3	93 ± 9.9	0	0

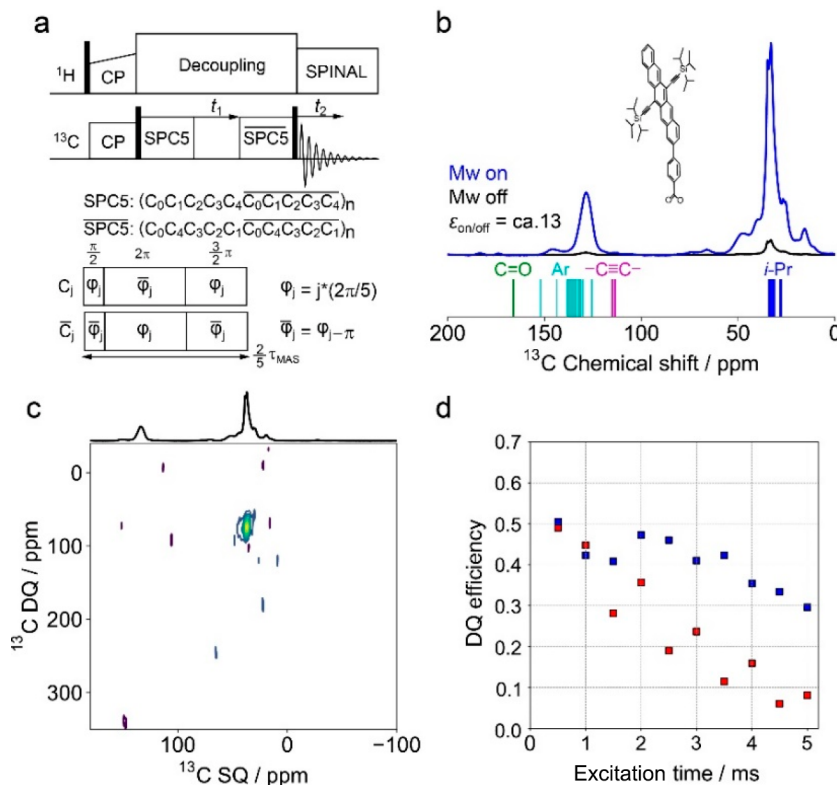


Figure 3. DNP-NMR measurements of CdTe-Pc. (a) SPC5 Pulse sequence for 2D ¹³C-¹³C DQ dipolar correlation spectra. (b) ¹³C CP/MAS spectra of CdTe-Pc (*n* = 26) with and without microwave irradiation. (c) 2D ¹³C-¹³C DQ dipolar correlation spectra of CdTe-Pc (*n* = 26). The spectrum was recorded using a DQ excitation time of 0.5 ms, 64 *t*₁ increments and 128 repetitions for each *t*₁ increment. The repetition delay was 4 s. (d) DQ buildup curves for CdTe-Pc (*n* = 26) (blue squares) and CdTe-Pc (*n* = 6.6) (red squares).

properties of SF are shown in Table 1 (see Sections 4 and 5 in the Supporting Information).

CdTe-Pc with a small number of Pc units (*n* = 6.6) has a surface Pc density of 0.31 nm⁻² and an average intermolecular distance of 1.8 nm. As shown in Table 1 and Figure S8, excitons generated in QDs successfully transferred to the singlet states of Pc with a quantum yield of 87 ± 5.9%. However, no triplet signal was observed from the TA spectra of CdTe-Pc (*n* = 6.6), suggesting that SF did not occur in the composite with the small *n* value. This finding could be ascribed to the intermolecular distance of 1.8 nm, which is too long to produce electronic coupling between surface Pc molecules. This result also confirmed that direct triplet energy transfer does not occur in the CdTe-Pc (*n* = 6.6) system. As the intermolecular distance decreased from 1.8 to 1.0 nm, the triplet yield reached 159 ± 6.8% for CdTe-Pc (*n* = 20).

To confirm the hypothesis that the intermolecular distance of Pc on a QD holds the key to SF, we further investigated the effect of the surface Pc density on SF by synthesizing CdTe@CdS core@shell. In the core@shell system, we can change only the surface area of QDs without significant changes in *n* or the optical properties of QDs. We realized a first absorption peak at 600 nm for the CdTe@CdS QDs similar to that of the CdTe

QDs, while the size of CdTe@CdS (5.4 nm) was larger than that of CdTe (3.5 nm). Although *n* in CdTe@CdS increased to 30, the surface Pc density decreased to 0.65 nm⁻², leading to a longer intermolecular distance (1.3 nm). FRET occurred from CdTe@CdS to the surface Pc, but no SF was observed, clearly demonstrating that the intermolecular distance is a key parameter to provide favorable intermolecular coupling for SF.

The NMR method can also provide information on ¹³C-¹³C correlation between Pc ligands by employing the SPC-5 homonuclear recoupling technique^{45,46} (Figure 3a), which can estimate dipole-dipole coupling between nuclei of interest. However, the intrinsic low sensitivity of NMR prevents the application of such measurements for natural-abundance samples. In this work, to enhance the sensitivity of NMR, we used a DNP method.⁴⁷⁻⁵² DNP-enhanced ¹³C cross-polarization (CP) magic angle spinning (MAS) spectrum of CdTe-Pc (*n* = 26) showed a ca. 13 times signal enhancement (Figure 3b).

On the basis of this result, 2D ¹³C-¹³C dipolar DQ correlation spectra were recorded using the homonuclear recoupling sequence (Figure 3c). For both samples, a correlation peak was observed at approximately 38 ppm, corresponding to ¹³C in CH₃ and CH groups in the isopropyl

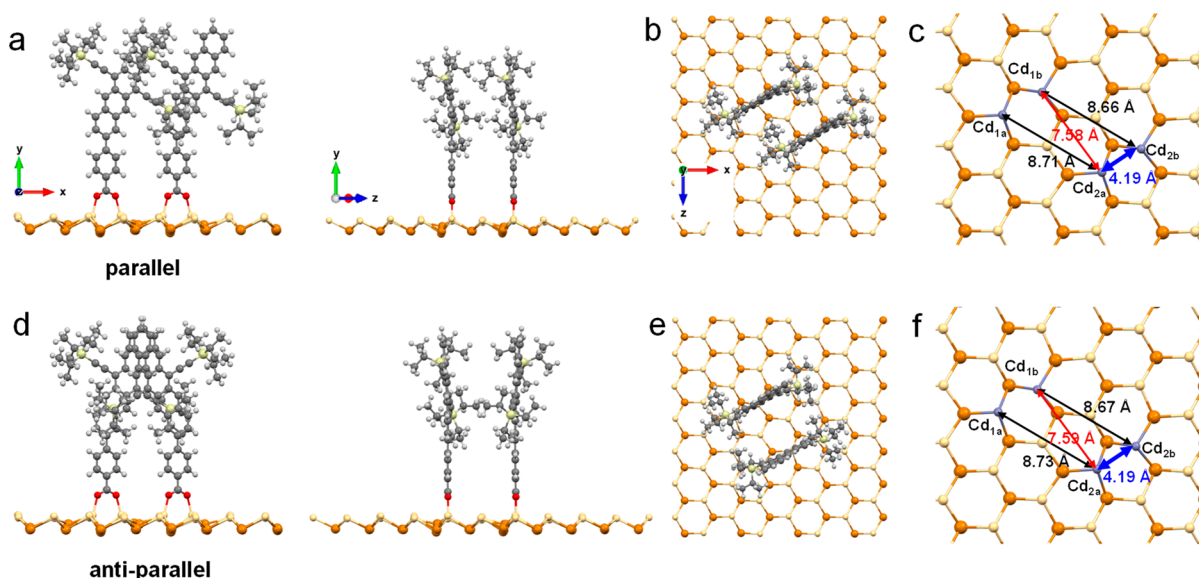


Figure 4. Computed molecular configurations of Pc on a CdTe surface. Parallel (a) and antiparallel (d) configurations of Pc dimer after geometrical optimization. The corresponding top views of the parallel and antiparallel configurations are given in (b) and (e), respectively. Atomic arrangements of CdTe(111) directly related to the intermolecular distance of Pc dimer with the parallel (c) and antiparallel (f) configurations. The black values correspond to the intermolecular direct distance between the Pc molecular faces, while the red values are orthogonal intermolecular distances between two parallel planes where each Pc molecular plane exists. The parallel deviation of paired Pc molecules (blue values) can be calculated through the Pythagorean theorem.

(*i*-Pr) units (Figure 3b), indicating that $\text{CH}_3\text{--CH}_3$ and $\text{CH}_3\text{--CH}$ are in close proximity. To obtain further insight, the DQ efficiency buildup curves were recorded (Figure 3d). At short excitation times, the DQ efficiencies were similar for both samples. The main contribution to the DQ efficiencies at short times is intramolecular $\text{CH}_3\text{--CH}_3$ and $\text{CH}_3\text{--CH}$ coupling in the *i*-Pr unit. The difference in DQ efficiencies for CdTe-Pc with $n = 26$ and 6.6 becomes larger with an increase in the excitation time, indicating the significant contribution of intermolecular dipole–dipole coupling for CdTe-Pc with a high n value of 26. The n dependence of the DQ efficiency buildup curve indicates that the observed signal corresponds to the intermolecular interaction between Pc and a QD. These results are in good agreement with the results of fsTA spectra. A computational simulation of the DQ buildup curve suggested that the major distance of intermolecular CH_3 in the *i*-Pr unit is around 4 Å (Figure S16).

The computational method predicted the most likely arrangement of Pc dimer on a CdTe surface. For the nearest-neighbor arrangement of the Pc dimer, we consider the parallel and antiparallel configurations of Pc on a CdTe(111) surface as a representative facet of QD (Figure 4a,b,d,e). From the results of the geometrical optimization, the two Pc units stably coordinated on the CdTe(111) surface in a parallel or antiparallel configuration. The intermolecular distance and shift length between Pc units are estimated to be ~ 8.7 and ~ 4.2 Å, respectively, in both configurations (Figure 4c,f). The intermolecular distance is 8.7 Å, which agrees with 9 Å estimated from the surface density of Pc, which realizes the present efficient SF. The deviation of Pc molecular faces, 4.2 Å, is larger than that reported in the crystalline solid of pentacene.²⁴ The unique molecular arrangement of the Pc pair was determined by the atomic arrangement of CdTe(111) because of the coordination between the carboxylic acid group and the Cd atom.

The computed total energy of the parallel configuration is 0.031 eV lower than that of the antiparallel configuration, indicating that the parallel configuration is more preferred than the antiparallel configuration. In fact, the intermolecular $\text{CH}_3\text{--CH}_3$ distance in the *i*-Pr unit indicated by DNP-NMR, ~ 4.0 Å, agreed well with the value of 4.5 Å estimated in the optimized parallel configuration (Figure 4a,b). Furthermore, the molecular orbital calculation indicated the partial intermolecular delocalization of molecular orbitals of Pc only in the parallel configuration, due to its larger overlapping area of Pc molecular face, which should be favorable for SF (Figure S17). Therefore, we conclude that an increase in n leads to the formation of the parallel Pc dimer resulting in a dramatic increase in SF efficiency.

The DNP-NMR and computational calculation indicated that the configuration of Pc is determined by the atomic arrangement of the CdTe(111) surface. The CdTe(111) surface structure promotes building the parallel and largely mismatched Pc dimer, which has not been reported in any conventional systems. The unique molecular arrangement, which was achieved due to the stable attachment of Pc pairs on the CdTe(111) surface, realized the current near-unity SF.

The arrangement of chromophores also plays an important role in controlling the dissociation and recombination of free triplets. An investigation of SF dimers demonstrated that strongly conjugated molecules led to radiative recombination and other competitive processes to free triplets, reducing the triplet yield and lifetime.²² Thus, a unity SF efficiency can be obtained by dimers, but the corresponding triplet lifetime is short because the diffusion of free triplets is prohibited in an SF system with a small number of molecules.¹⁷ Achieving balanced intermolecular coupling is the key to efficient SF and is challenging. Nanosecond transient absorption (nsTA) spectra show that $\text{Pc}(T_1)$ generated on the surface of CdTe QDs has a long lifetime (~ 30 μs) (Figure S15) that is longer than the previously reported values for crystalline solids and

covalent dimers of pentacene.^{17,53} From the result of a kinetic analysis of the decay profile of $\text{Pc}(T_1)$, the decay corresponding to the back-reaction (i.e., triplet–triplet annihilation) was negligible (0.2%). The estimated triplet yield of SF from the long-lived component of ns-TAS was 184%, indicating that almost no back-reaction occurred even in the ms region. Since no signals of Pc aggregation were observed in the spectroscopic investigation, we believe that the interaction between Pc molecules is negligible on the QD surface.¹³ Such a nonconjugated arrangement of Pc on the surface of CdTe provides a through-space interaction for SF, which will prevent undesired deactivation processes. The Pc units on QDs exquisitely gather for the efficient dissociation and maintain of free triplets.

CONCLUSION

We have demonstrated that the composition of QDs and molecules provides an important option for designing ideal SF systems. A CdTe QD is an excellent scaffold from the viewpoint of compensation of the weak light-harvesting ability of Pc and specific arrangements of surface-attached Pc molecules in improving the triplet yield and lifetime of SF.

The unique properties of Pc configured on the nano-interfaces and synergy with QDs could expand the possibilities of SF for a variety of applications, including minuscule optical devices, energy devices, and photoenergy conversion.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.1c04731>.

Experimental procedures, figures, and corresponding discussions of additional supporting experimental data (PDF)

AUTHOR INFORMATION

Corresponding Authors

Masanori Sakamoto – Institute for Chemical Research, Kyoto University, Uji, Kyoto 611-0011, Japan; orcid.org/0000-0001-5018-5590; Email: sakamoto@scl.kyoto-u.ac.jp

Hayato Sakai – Department of Chemistry, Faculty of Science and Technology, Keio University, Yokohama, Kanagawa 223-8522, Japan; Email: sakai@chem.keio.ac.jp

Toshiharu Teranishi – Institute for Chemical Research, Kyoto University, Uji, Kyoto 611-0011, Japan; orcid.org/0000-0002-5818-8865; Email: teranishi@scl.kyoto-u.ac.jp

Authors

Jie Zhang – Department of Chemistry, Graduate School of Science, Kyoto University, Uji, Kyoto 611-0011, Japan; orcid.org/0000-0002-3079-0977

Katsuaki Suzuki – Institute for Chemical Research, Kyoto University, Uji, Kyoto 611-0011, Japan

Taku Hasobe – Department of Chemistry, Faculty of Science and Technology, Keio University, Yokohama, Kanagawa 223-8522, Japan; orcid.org/0000-0002-4728-9767

Nikolai V. Tkachenko – Chemistry and Advanced Materials Group, Faculty of Engineering and Natural Sciences, Tampere University, FI33720 Tampere, Finland; orcid.org/0000-0002-8504-2335

I-Ya Chang – Department of Chemistry, Kyoto University, Kyoto, Kyoto 606-8502, Japan

Kim Hyeon-Deuk – Department of Chemistry, Kyoto University, Kyoto, Kyoto 606-8502, Japan; orcid.org/0000-0002-5815-8041

Hironori Kaji – Institute for Chemical Research, Kyoto University, Uji, Kyoto 611-0011, Japan; orcid.org/0000-0002-5111-3852

Complete contact information is available at: <https://pubs.acs.org/doi/10.1021/jacs.1c04731>

Notes

The authors declare no competing financial interest.

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